

Student Resource Guide

Generative AI for Beginners

Section 4: Image Generation AI | Lecture 4.6: GANs - An Alternative Approach

Learning Objectives

- Explain the adversarial training dynamic between the GAN generator and discriminator
- Describe the training challenges of GANs including mode collapse and vanishing gradients
- Compare GAN capabilities and limitations against diffusion models across specific use cases
- Identify contemporary applications where GANs remain the preferred technical approach
- Evaluate the historical contribution of GANs to the development of generative AI

Introduction

Generative Adversarial Networks were the first practical approach to high-quality image synthesis and dominated the field from 2014 through 2022. Understanding GANs is not merely historical—they continue to be used in applications where their specific characteristics (fast inference, high resolution, training on limited data) make them preferable to diffusion models. Moreover, GANs introduced concepts—the generator-discriminator framework, adversarial training dynamics, the latent vector as a controllable parameter—that remain foundational to generative AI thinking.

The GAN framework also introduced the field to some of its most challenging training problems: mode collapse (where the generator learns to produce only a narrow range of outputs), vanishing gradients (where the discriminator becomes too powerful and stops providing useful training signal), and the brittleness of equilibrium training. Understanding these failure modes illuminates why the field moved toward diffusion models, while also explaining the specific niches where GANs remain the better tool.

For practitioners, this lecture provides the conceptual vocabulary for evaluating generative model trade-offs. The comparison between GANs and diffusion models—fast vs. iterative inference, sharp vs. high-fidelity quality, limited vs. broad controllability—is a template for how to reason about any two generative approaches.

Key Concepts

GAN Training Dynamic (Adversarial Competition)

Definition	A GAN consists of a generator network (G) that maps random noise vectors to images, and a discriminator network (D) that classifies images as real (from training data) or fake (from G). Both are trained simultaneously with opposing objectives: D maximises classification accuracy; G maximises its ability to fool D. At training equilibrium, G produces images indistinguishable from real ones.
Business Context	The adversarial dynamic creates an automatic quality standard: the generator must reach the quality threshold of 'fooling a trained classifier'. This is self-supervised—no manual labelling of generated images is required. However, the competitive training dynamic is inherently unstable, requiring careful hyperparameter tuning.
AI Practitioner relevance	Practitioners using GAN-based tools should understand that training GANs is significantly more engineering-intensive than fine-tuning diffusion models. Pre-trained GANs (StyleGAN3, GigaGAN) are practical to use but difficult to retrain; domain adaptation is challenging compared to diffusion model LoRA fine-tuning.
Real-World Example	NVIDIA's StyleGAN2 was trained on the FFHQ (Flickr Faces High Quality) dataset—70,000 high-resolution face photographs. The resulting model generates photorealistic human faces at 1024×1024 resolution. The generator never saw these faces; it learned to produce statistically indistinguishable new faces through adversarial training.

Mode Collapse

Definition	A training failure in GANs where the generator learns to produce a very narrow range of outputs—often a single or few image types—that consistently fool the current discriminator, rather than learning the full diversity of the training
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	distribution.
Business Context	Mode collapse makes GAN outputs commercially useless for applications requiring diversity—you receive many near-identical outputs instead of varied content. Detection requires monitoring output diversity metrics (FID, Inception Score) during training; prevention requires training stabilisation techniques (gradient penalty, spectral normalisation, minibatch discrimination).
AI Practitioner relevance	Practitioners evaluating GAN outputs should test diversity explicitly: generate 50–100 images from random seeds and verify meaningful visual diversity. If outputs cluster into a few visual types, the model has partially collapsed. This problem does not exist in the same form for diffusion models, which have inherent stochasticity through the noise initialisation.
Real-World Example	A GAN trained to generate diverse product images on a small dataset began producing only images with blue backgrounds—the most common colour in the training set. The discriminator had become good at rejecting non-blue products as 'fake', causing the generator to collapse into blue-background production.
Key Risks	Partial mode collapse can be subtle—outputs look diverse on casual inspection but cluster in specific attribute spaces (colour, composition, expression). Always use quantitative diversity metrics, not just visual spot-checks.

Latent Space Control in GANs

Definition	In GANs, the generator maps random latent vectors (z) to images. The latent space is unstructured by default, but can be made more interpretable through architectural choices. StyleGAN's $W+$ space maps z through a mapping network to an intermediate latent space where individual dimensions correspond more interpretably to visual attributes.
Business Context	Controllable latent spaces enable attribute-targeted generation: 'generate face images

	with age $\in [30-35]$, without glasses, frontal pose'. This is valuable for diverse training data generation, personalised avatar creation, and visual content auditing.
AI Practitioner relevance	When using GAN-based tools that expose latent controls (e.g., StyleGAN interfaces), practitioners can perform latent arithmetic: encode a real face into the latent space, shift the latent vector in the 'smiling' direction, decode to produce a smiling version. These operations have no direct equivalent in prompt-only diffusion workflows.
Real-World Example	InsightFace's reconstruction tools use StyleGAN's controllable latent space to perform age progression on portraits: by moving the latent vector along a learned 'age direction', the tool generates convincing older versions of input faces while preserving identity—used in forensic age progression and entertainment.

GAN vs. Diffusion Model Trade-offs

Definition	Key differences between GANs and diffusion models across dimensions including inference speed, output quality, text conditioning capability, training stability, and ease of fine-tuning for new domains.
Business Context	Tool selection should match use case: GANs for real-time applications requiring single-pass generation (gaming NPC appearance, live video filters); diffusion models for high-quality, text-conditioned generation where latency is acceptable (content creation, concept design, stock imagery).
AI Practitioner relevance	Practitioners should benchmark both approaches on their specific use case rather than defaulting to the most popular tool. For streaming video generation, game character creation, and applications where a pre-trained GAN already covers the required visual domain, GANs may provide better economics.
Real-World Example	Snapchat's real-time face filter technology uses GAN-based face synthesis for sub-30ms latency generation—impossible with diffusion models requiring 20+ inference

	steps. The quality is lower than diffusion models, but real-time performance is the non-negotiable requirement.
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Real-World Applications

Despite diffusion models' dominance in general image generation, GANs retain competitive advantages in specific application categories.

Real-Time Video and Image Filters: GAN inference is a single forward pass, achieving sub-20ms latency on GPU hardware. This makes GANs the only practical approach for real-time face filters, video style transfer, and augmented reality applications where diffusion models' multi-step inference is prohibitively slow.

Super-Resolution and Image Enhancement: SRGAN and ESRGAN (Enhanced SRGAN) use discriminator training to produce sharper, more realistic textures in upscaled images than traditional interpolation methods. Many production super-resolution tools (used in streaming quality enhancement, legacy photo restoration) use GAN-based architectures for their superior detail generation.

Deepfake and Face Synthesis: Early deepfake technology and many current face-swapping tools use GAN-based autoencoder architectures that share encoders between source and target identities. The single-pass inference is critical for real-time or near-real-time face synthesis in video production.

Synthetic Training Data Generation at Scale: When a specific visual domain requires training data (defect images, rare medical conditions, edge-case driving scenarios), a GAN trained on small existing datasets can be used to generate large synthetic datasets quickly. Single-pass generation at scale is dramatically faster than diffusion models for this use case.

Best Practices

Use pre-trained GANs rather than training from scratch when possible. Training a GAN from scratch requires careful hyperparameter tuning, training stability techniques, and significant compute. Pre-trained models (StyleGAN3, ESRGAN) for common domains are reliable and accessible.

Monitor output diversity quantitatively with FID scores. Frechet Inception Distance (FID) measures the statistical similarity between generated and real image distributions—lower is better. Track FID throughout training to detect mode collapse before it becomes severe.

Use gradient penalty (WGAN-GP) training for improved stability. Vanilla GAN training with binary cross-entropy loss is prone to discriminator overpowering and vanishing gradients. WGAN with gradient penalty provides much more stable training dynamics.

For fine-tuning GANs on new domains, use transfer learning from a large pre-trained model. Fine-tuning a pre-trained StyleGAN2 on a small domain-specific dataset (few hundred images) often produces better results than training from scratch, and requires significantly less data and compute.

Common Mistakes to Avoid

Mistake: Assuming GANs are obsolete and can be ignored because diffusion models dominate news coverage.

Correct approach: GANs remain the better tool for real-time inference, single-image super-resolution, and specific structured generation tasks. Technology selection should be use-case-driven. Stay current on GAN developments as well as diffusion model advances.

Mistake: Evaluating GAN output quality using only visual spot-checks of a few samples.

Correct approach: Mode collapse and distribution coverage failures are not always visible in individual samples. Use quantitative diversity metrics (FID, IS, precision/recall) and generate large batches (100+ images) to assess coverage of the full target distribution.

Mistake: Treating GAN latent space navigation as equivalent to diffusion model prompt engineering.

Correct approach: GAN latent space requires different tools (latent encoding, attribute vector arithmetic, interpolation) and has different failure modes. Prompting skills from diffusion models do not directly transfer; learn the latent space manipulation interface specific to the GAN tool being used.

Practitioner Tool: GAN vs. Diffusion Model Selection Guide

Type: Decision Framework

Use this framework to select the appropriate generative approach for a new image generation use case.

Decision Criterion	Choose GAN	Choose Diffusion Model
Inference latency requirement	< 50ms (real-time)	> 1 second acceptable
Text-prompt conditioning needed?	No (or minimal)	Yes (primary interface)
Training data availability	Small domain dataset (100s-1000s)	Large diverse dataset or pre-trained
Output diversity required?	Monitor carefully (collapse risk)	Inherently high diversity
Super-resolution /	Prefer (ESRGAN, Real-	Less common

enhancement?	ESRGAN)	
Fine-tuning complexity tolerance	High (requires GAN expertise)	Low (LoRA, DreamBooth)
Production deployment maturity	Established tooling	Rapidly maturing APIs
Quality ceiling for complex scenes	Lower	Higher

Note: If a use case falls into both columns across different criteria, prototype both approaches and benchmark on your specific quality and latency metrics before committing.

Knowledge Check

Question 1: A company needs to generate synthetic face images for a training dataset at high throughput—approximately 100,000 images—with single-GPU compute. Which approach is most appropriate?

- A. Diffusion model at 50 inference steps per image
- **B. Pre-trained GAN (e.g., StyleGAN3) with single-pass inference**
- C. Text-to-image API with 100,000 individual API calls
- D. Manual photography and labelling

Correct Answer: B. Pre-trained GAN (e.g., StyleGAN3) with single-pass inference

Explanation: Generating 100,000 images with a diffusion model at 50 steps requires 5 million forward passes—extremely slow. A GAN generates each image in a single forward pass, making high-throughput synthetic data generation practical on single-GPU hardware.

Question 2: A team training a GAN on product images notices that after 50 epochs, all generated images have similar blue-toned backgrounds despite the training data having diverse backgrounds. What has most likely occurred?

- A. The learning rate is too low
- B. The discriminator has been overtrained
- **C. Mode collapse—the generator has converged to a narrow output distribution**
- D. The VAE encoder is not functioning correctly

Correct Answer: C. Mode collapse—the generator has converged to a narrow output distribution

Explanation: Mode collapse is the characteristic GAN failure where the generator produces outputs of narrow diversity that fool the current discriminator, rather than covering the full training distribution. The discriminator found blue backgrounds easy to pass, causing the generator to default to them.

Question 3: Snapchat uses real-time face filters that apply style transformations at under 30ms latency on a mobile device. Why is this application better suited to a GAN than a diffusion model?

- A. GANs produce higher visual quality than diffusion models at any resolution
- B. Diffusion models cannot process face images
- **C. GAN inference is a single forward pass, enabling sub-30ms latency; diffusion models require 20+ iterative denoising steps**
- D. Snapchat's training data is too small for diffusion models

Correct Answer: C. GAN inference is a single forward pass, enabling sub-30ms latency; diffusion models require 20+ iterative denoising steps

Explanation: Real-time video filter performance requires sub-30ms latency—achievable with a single-pass GAN forward pass but not with diffusion models that require 20+ sequential denoising steps even with accelerated samplers.

Reflection Questions

Take time to consider the following questions before the next session:

1. GANs solved a critical problem—training two networks with opposing objectives—using only standard gradient descent. What does this achievement say about the power of framing problems in terms of optimisation objectives?
2. Mode collapse is an emergent failure: the generator 'finds' a shortcut that technically satisfies the training objective but misses the intent. Where else in AI development do you see systems finding unintended shortcuts?
3. The GAN discriminator can be repurposed as a quality classifier after training. What other components of deep learning systems might be reusable in ways beyond their original training purpose?
4. GANs were state-of-the-art for 8 years before being largely displaced by diffusion models. What lesson does this cycle suggest about investing in and building on any current best-practice architecture?
5. Given that GANs are faster at inference and diffusion models are better at diversity and text conditioning, what future architecture might combine the best of both approaches?

Key Takeaways

- GANs use an adversarial generator-discriminator framework: the generator creates fake images; the discriminator learns to detect them; competition drives quality.

- Mode collapse—where the generator produces only narrow output varieties—is the characteristic GAN training failure, detectable through diversity metrics like FID.
- GANs produce images in a single forward pass, making them the only viable approach for real-time applications (video filters, streaming AR) requiring sub-50ms latency.
- StyleGAN's mapping network creates a structured latent space ($W+$) enabling direct attribute control through latent vector arithmetic.
- GAN super-resolution (ESRGAN) remains competitive for image enhancement tasks even as diffusion models dominate general generation.
- Tool selection between GANs and diffusion models should be use-case-driven: latency, text conditioning, diversity requirements, and fine-tuning complexity are the key decision criteria.

Next Actions

Complete at least two of the following before the next lecture:

6. Visit thispersondoesnotexist.com (StyleGAN-generated faces) and generate 20 images—inspect for GAN artifacts (ear asymmetry, background texture inconsistencies near face boundaries) to calibrate your detection skills.
7. Install Real-ESRGAN and apply it to a low-resolution image from your work domain—compare the output quality to bicubic upscaling to understand the practical benefit.
8. Research the FID (Frechet Inception Distance) metric; understand what it measures, how it is calculated, and why it is the standard benchmark for evaluating generative image model quality.
9. Compare a StyleGAN3 interface (e.g., via Hugging Face demos) to a Stable Diffusion interface side-by-side—note the difference in control paradigm (latent sliders vs. text prompts) and output character.